

PiXL KnowIT!

GCSE Biology

AQA Topic – Infection and Response

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Infection and Response

Communicable diseases

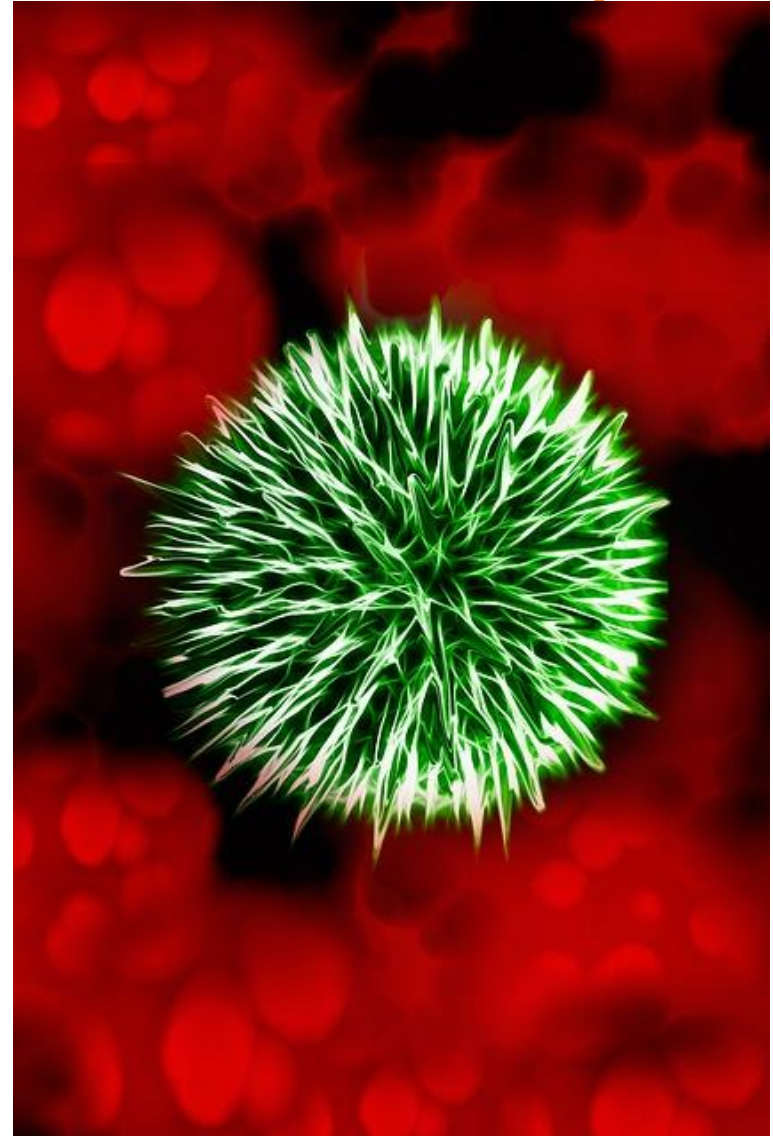
- Communicable (infectious) diseases
- Viral diseases
- Bacterial diseases
- Fungal diseases
- Protist diseases
- Human defence systems
- Vaccination
- Antibiotics and painkillers
- Discovery and development of drugs

Monoclonal Antibodies (Biology HT only)

- Producing monoclonal antibodies
- Uses of monoclonal antibodies

Plant Disease (Biology only)

- Detection & identification of plant diseases
- Plant defence responses



Infection and Response

Part 1

- **Communicable**
- **Viral diseases**
- **Bacterial disease**
- **Fungal disease**
- **Protist diseases**



Pathogens are **micro organisms** that cause **infectious disease**.

Pathogens may **infect plants or animals** and can be **spread** by **direct contact**, by **water** or by **air**.

Pathogens **depend** on the **host** to provide the suitable **conditions and nutrients** that they need to **grow and reproduce**.

Pathogens can be **bacteria, viruses, fungi** or **protists**.

Viral diseases

- **Viruses may reproduce rapidly**
- **Viruses live and reproduce inside cells causing damage**

Bacterial diseases

- **Bacteria may reproduce rapidly**
- **Bacteria may produce toxins that damage tissues and make us feel ill**

Pathogens may **infect plants or animals** and can be **spread** by **direct contact (D)**, by **water (W)** or **by air (A)**.



Examples of Viral Diseases

Measles

- Symptoms of fever and a **red skin rash**
- Can be **fatal** if complications occur
- Spread by **inhaling** droplets containing the virus from sneezes and coughs
- It is controlled by vaccinating young children



HIV

- Initially causes a **flu-like** illness and spread by **sexual contact** or exchange of body fluids such as blood when drug users share needles
- Unless HIV is successfully controlled with **antiretroviral** drugs, the virus will attack the body's immune cells
- Late stage HIV or **AIDS** occurs when the body's immune system can no longer deal with other infections or cancers



Tobacco Mosaic virus

- Common plant virus which enters through a damaged epidermis
- Seen as a distinct **mosaic** discolouration pattern on the leaves
- Affects **growth** as **photosynthesis** cannot occur as efficiently
- Control by removing affected leaves and destroying pests which caused initial epidermal damage



Examples of Bacterial Diseases

Salmonella

- Salmonella food poisoning is spread by bacteria **ingested** in food or on food prepared in **unhygienic** conditions.
- Bacteria secrete **toxins** and cause symptoms including fever, abdominal cramps, diarrhoea and **vomiting**.
- In the UK, **poultry** are **vaccinated** against salmonella to **control** the spread of the disease.



Gonorrhoea

- Sexually transmitted disease (**STD**) caused by bacteria.
- Causes a thick yellow or **green discharge** from the **penis** or **vagina** and pain when urinating.
- Can be controlled with **antibiotics** or barrier methods of contraception such as a **condom**.
- Easily treated with the antibiotic **penicillin** until many **resistant bacterial** strains were found.



Example of a Fungal Disease

Rose Black Spot



- Purple or black spots develop on leaves
- Leaves often turn yellow and drop off
- **Photosynthesis** is reduced. Growth is affected as a result
- Fungal **spores spread** by wind or water
- Treat by **removing** infected leaves and burning them
- Spray with **fungicide** (a pesticide which is used to kill fungus)

Example of a Protist Disease

Malaria

- The malaria causing **protist** is spread by **mosquitoes** feeding on infected blood and then biting a human
- Mosquitoes are **vectors** as they pass on malaria but do not suffer themselves
- Symptoms include **recurrent fever** and malaria can be fatal
- Control the **spread** by preventing mosquitoes **breeding** and use mosquito **nets** to avoid being bitten



Infection and Response part 1 – Types of diseases

Disease	Symptom	Method of transmission	Control spread of disease by:	Caused by:
Malaria	Recurrent fever	By a vector from an infected person	Preventing breeding of mosquitoes or use of a net to prevent being bitten	Protist
Tobacco Mosaic Virus	Mosaic pattern on leaves	Enters via wounds in epidermis caused by pests	Remove infected leaves and control pests which are damaging leaves	Virus
Measles	Fever Red skin rash	Droplet infection from sneezes and coughs	Child Vaccination	Virus
Gonorrhoea	Green discharge from penis or vagina	Direct sexual contact or body fluids	Use of a condom and treat infected person with antibiotics	Bacteria
Rose Black Spot	Purple black spots on leaves	Spores carried via wind or water	Remove infected leaves and spray with pesticide	Fungus
Salmonella	Fever, cramp, vomiting, diarrhoea	Food prepared in unhygienic conditions or not cooked properly	Improve food hygiene, wash hands, vaccinate poultry, cook food thoroughly	Bacteria

QuestionIT

Communicable diseases Part 1a

- Viral diseases
- Bacterial diseases
- Fungal diseases
- Protist diseases



1. What is the definition of a pathogen?
2. List four types of micro-organism which can act as pathogens.
3. Copy and complete the table to describe one similarity and one difference between how bacteria and viruses act as pathogens.

Pathogen	Similarity	Difference
Bacteria		
Virus		

4. What do pathogens need from the host organism?
5. Complete the sentences:
 HIV can be successfully controlled with _____ drugs. If the immune system is badly damaged then _____ may develop.

6. Copy and complete the table for each disease.

Disease	Symptom	Method of transmission	Control spread of disease by:	Caused by:
Measles	Fever Red skin rash		Child Vaccination	
	Fever, cramp, vomiting, diarrhoea	Food prepared in unhygienic conditions or not cooked properly		Bacteria
Rose Black Spot	Purple black spots on leaves		Remove infected leaves and spray with pesticide	
Gonorrhoea		Direct sexual contact or body fluid exchange	Use of a condom and treat infected person with antibiotics	Bacteria
	Recurrent fever	By a vector from an infected person	Preventing breeding of mosquitoes or use of a net to prevent being bitten	
Tobacco Mosaic Virus	Mosaic pattern on leaves			Virus

7. Look at this photograph.

Suggest how pathogens could be transferred in this situation.

What could people do to reduce the spread of pathogens?



AnswerIT!

Communicable diseases Part 1

- Viral diseases
- Bacterial diseases
- Fungal diseases
- Protist diseases



1. What is the definition of a pathogen?

Micro-organisms which cause infectious disease in animals & plants.

2. List four types of micro-organism which can act as pathogens.

Bacteria, Virus, Protist, Fungus.

3. Copy and complete the table to describe one similarity and one difference between how bacteria and viruses act as pathogens.

Pathogen	Similarity	Difference
Bacteria	Reproduce rapidly inside the body	Produce toxins that damage tissues
Virus	Reproduce rapidly inside the body	Live and reproduce inside cells causing cell damage

4. What do pathogens need from the host organism?

Suitable conditions and nutrition to be able to grow and reproduce.

5. Complete the sentences:

HIV can be successfully controlled with antiretroviral drugs. If the immune system is badly damaged then AIDS may develop.

6. Copy and complete the table for each disease.

Disease	Symptom	Method of transmission	Control spread by:	Caused by:
Measles	Fever Red skin rash	Droplet infection from sneezes and coughs	Child Vaccination	Virus
Salmonella	Fever, cramp, vomiting, diarrhoea	Food prepared in unhygienic conditions or not cooked properly	Improve food hygiene, wash hands, vaccinate poultry, cook food thoroughly	Bacteria
Rose Black Spot	Purple black spots on leaves	Spores carried via wind/water	Remove infected leaves and spray with pesticide	Fungus
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Malaria	Recurrent fever	By a vector from an infected person	Preventing breeding of mosquitoes or use of a net to prevent being bitten	Protist
Tobacco Mosaic Virus	Mosaic pattern on leaves	wounds in epidermis caused by pests	Remove infected leaves and control pests which are damaging leaves	Virus

7. Look at this photograph.

Suggest how pathogens could be transferred in this situation.

What could people do to reduce the spread of pathogens?

Direct contact by touching a contaminated surface.

Droplet infection if someone sneezes or coughs in the lift.

Air - if fungal spores are present.



Could reduce the spread by:

Hand over mouth if coughing and then washing.

Use a tissue if sneezing and then dispose of it and wash hands.

Washing hands with soap after visiting the toilet.

Wear a face mask.

Infection and Response

Part 1b

-
- A hand holding a sphere covered in educational and learning-related terms. The word 'learning' is prominently displayed in large blue letters. Other visible words include 'knowledge', 'experience', 'search', 'strategy', 'innovation', 'school', 'science', 'knowledge', 'learning', 'inspiration', 'business', 'learning', 'inspiration', 'business', 'learning', 'inspiration'. The sphere is held by a hand, and the background is white.



n
o
s
e

Nasal hairs, sticky mucus and cilia prevent pathogens entering through the nostrils.



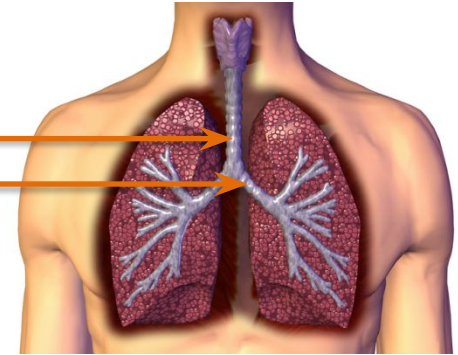
stomach

Stomach acid (pH1) kills most ingested pathogens

The human body has several non specific ways of defending itself from pathogens getting in.

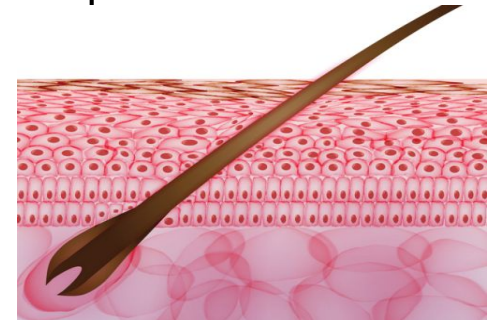
video

trachea
bronchus



Respiratory system is lined with mucus to trap dust and pathogens. Cilia move the mucus upwards to be

ski
n



Hard to penetrate waterproof barrier. Glands secrete oil which kill microbes.

Sometimes pathogens gain entry to the body.

The **immune system** takes over to destroy them.

The **white blood cells** are part of the immune system. **Pathogens** are identified by white blood cells because they have different surface proteins. We call these surface proteins **antigens**.

White blood cells act in **3** ways to **defend** the body:

1. White blood cells (called **phagocytes**) engulf the pathogens and digest them. This is called **phagocytosis**. [Video phagocytes](#)
2. White blood cells (called **lymphocytes**) identify the **antigen** on the pathogen. They make **specific antibodies** to destroy the pathogens. This can take time and so an infection may occur. If a person is **infected again** by the same pathogen, the white blood cells make the **antibodies** much **faster**. [Video lymphocytes](#)
3. Bacteria may produce toxin (poison). White blood cells release specific **antitoxins** to **neutralise** the effect of the toxin.

Communicable diseases can be dangerous leading to epidemics or pandemics. **Vaccination** can be used to **enhance** the immune system to reduce the chances of this happening. A vaccine contains a small amount of **dead or inactive** form of the pathogen that can be introduced into the body.



Lymphocyte



pathogen



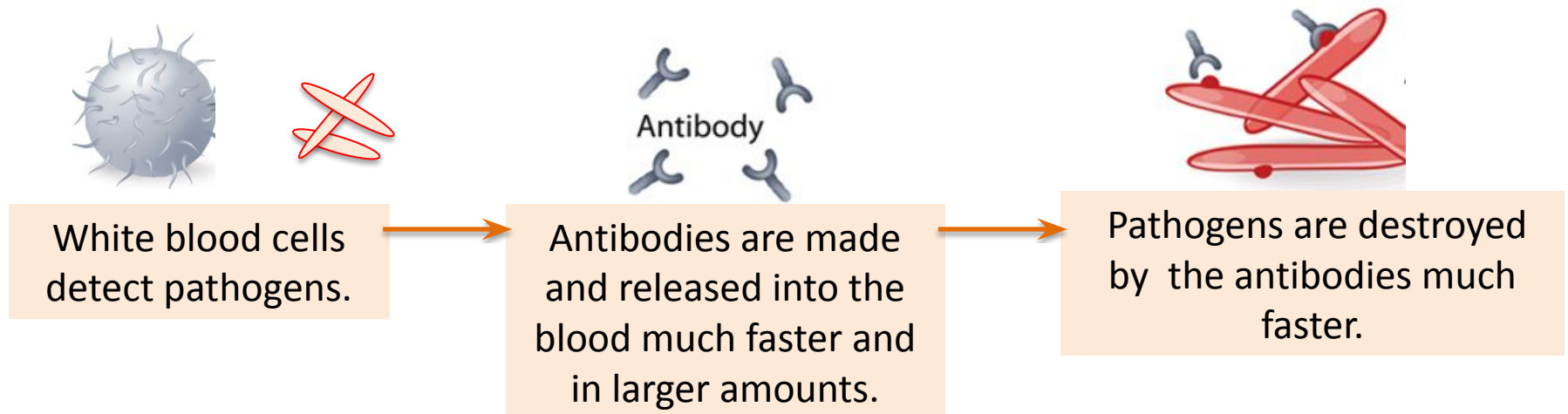
White blood cells
detect pathogens in
the vaccine.

Antibodies are released
into the blood.

Pathogens are destroyed by
antibodies.

If the body becomes **re-infected** with the same pathogen then the white blood cells are prepared.

The white blood cells can respond much more **quickly** and make **more** of the right type of **antibodies** much more quickly.



This means that the person is **unlikely to suffer** the symptoms of the harmful disease.

Infection has been prevented by **enhancing** the immune system.

An **antibiotic** is a drug that helps to cure a bacterial disease by killing the infective bacteria **inside** the body.

Different bacterial infections need a **different** antibiotic.



Antibiotics **cannot** be used to **treat viral** pathogens.

Penicillin is a well known antibiotic medicine.

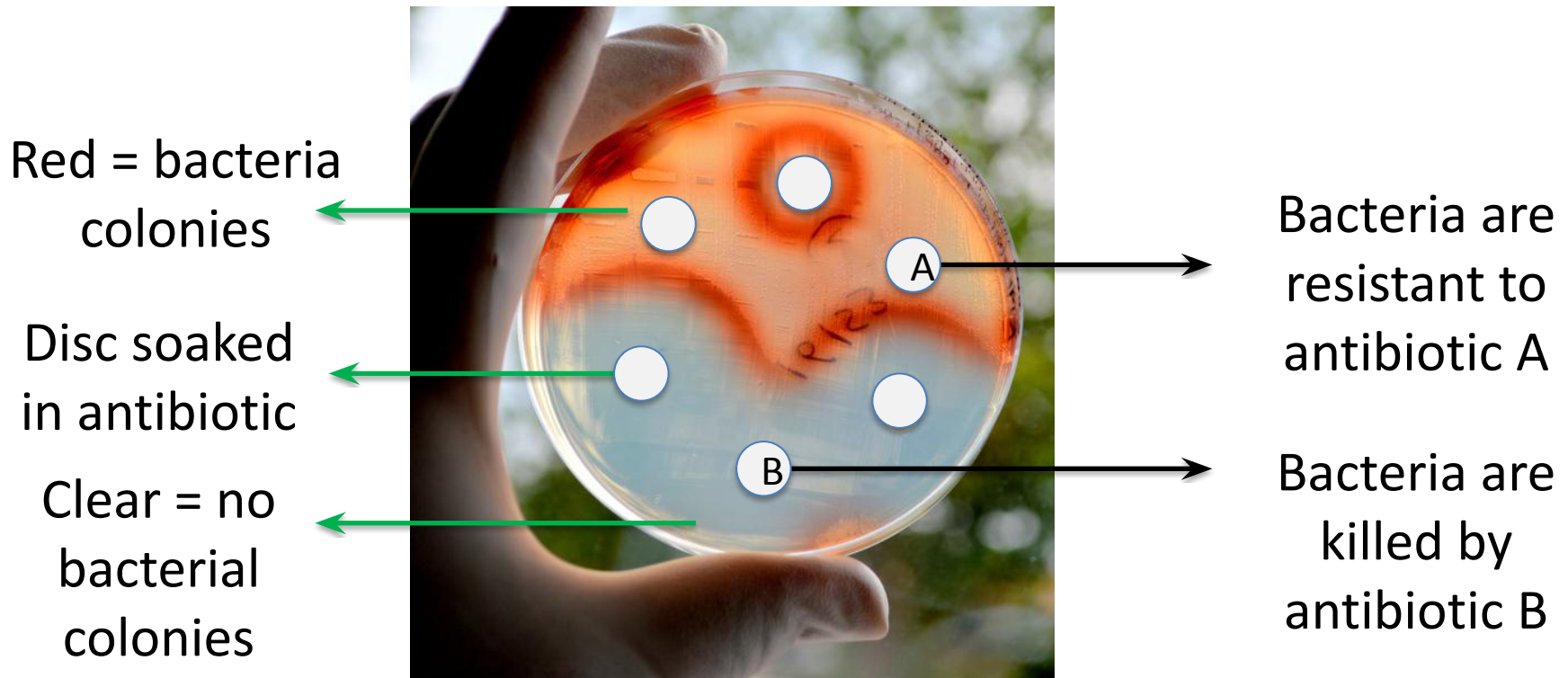
Using antibiotics has greatly **reduced** deaths.

It is difficult to develop **drugs** to kill **viruses** without harming body tissues because viruses live and reproduce inside cells.

Bacteria can **mutate**.

Sometimes this makes them **resistant** to antibiotic drugs.

The **mutated** bacteria are **not killed**. Increasing numbers of different bacteria are becoming resistant and this is **of concern**.



A doctor will **not prescribe** antibiotics for a **viral infection** as they do not work.

Antibiotics can only be used for **bacterial infections**

Painkillers, steroids or anti inflammatory medicines can be used to **relieve** the **symptoms** of viral infections. Symptoms may include: fever, muscle ache, headache or a runny nose.



Painkillers **do not** kill pathogens.

QuestionIT

Infection and Respons Part 1b

- Human defence systems
- Vaccination
- Antibiotics and painkillers



1. Name and describe 3 ways the human body defends against the entry of pathogens.
2. Draw a series of pictures to show how white blood cells carry out phagocytosis.
3. What is an antigen?
4. Why will the antibody produced for measles not be effective in other diseases?
5. Why are dead or inactive pathogens used in a vaccine?
6. What happens if the live pathogen invades the body after being vaccinated?
7. What is an antibiotic and what is it used for?

1. Name and describe 3 ways the human body defends against the entry of pathogens. **Skin – barrier, nose - nasal hairs , mucus and cilia, trachea & bronchi – mucus to catch, cilia to remove from lungs, stomach - HCl acid kills.**
2. Draw a series of pictures to show how white blood cells carry out phagocytosis. **Diagram of white blood cell moving towards and engulfing foreign body and then adding digestive enzymes.**
3. What is an antigen? **A protein on the surface of a pathogen.**
4. Why will the antibody produced for measles not be effective in other diseases? **Antibody has a specific shape which fits with a specific antigen.**

5. Why are dead or inactive pathogens used in a vaccine?

To stimulate white blood cells to make antibodies without causing the disease.

6. What happens if the live pathogen invades the body after being vaccinated?

White blood cells able to respond much faster and produce more antibodies quicker to destroy the pathogens so the person doesn't suffer the symptoms of the disease.

7. What is an antibiotic and what is it used for?

Kills bacteria inside the body without harming human cells.

8. Why is there a growing concern about bacterial resistance to antibiotics?

9. Which one of these statements is true?

Painkillers are used to treat the symptoms of a disease and kill the pathogens which cause it.

Painkillers are used to kill the pathogens causing a disease.

Painkillers are used to treat the symptoms of a disease but do not kill the pathogens.

AnswerIT!

Infection and Response Part 1b

- Human defence systems
- Vaccination
- Antibiotics and painkillers



8. Why is there a growing concern about bacterial resistance to antibiotics? **The resistant bacteria are not killed and can continue to multiply inside the body making the person very ill and infecting others too.**

9. Which one of these statements is true?

Painkillers are used to treat the symptoms of a disease and kill the pathogens which cause it

Painkillers are used to kill the pathogens causing a disease

Painkillers are used to treat the symptoms of a disease but do not kill the pathogens.

Infection and Response

Part 2

-
- A hand holding a sphere covered in educational and learning-related terms like 'learning', 'knowledge', 'experience', 'search', 'strategy', 'innovation', 'school', 'science', 'knowledge', 'learning', 'inspiration', 'business', 'learning', 'inspiration'.

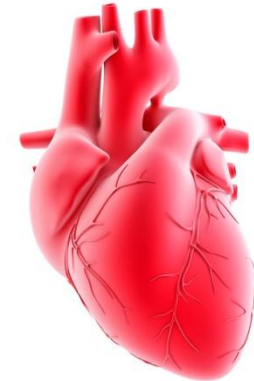
Traditionally drugs were extracted from plants and micro-organisms.
video



Aspirin is a painkiller and anti-inflammatory drug. This was first found in **willow bark**.



Foxglove plants have been a source of the drug digitalis which acts on the **heart**.



Alexander Fleming is famous for discovering ***Penicillium*** mould.



Fleming noticed the fungus on unwashed equipment and went on to discover its antibiotic properties.

Most **new drugs** are **developed** and **synthesised (made)** in a laboratory by chemists in the pharmaceutical industry. The initial chemical may have been sourced from a plant.

New drugs must be tested and trialled to check:

- ✓ **Efficacy** - that the drugs work
- ✓ **Toxicity** - that the drug is not poisonous
- ✓ **Dose** - the most suitable **amount** to take

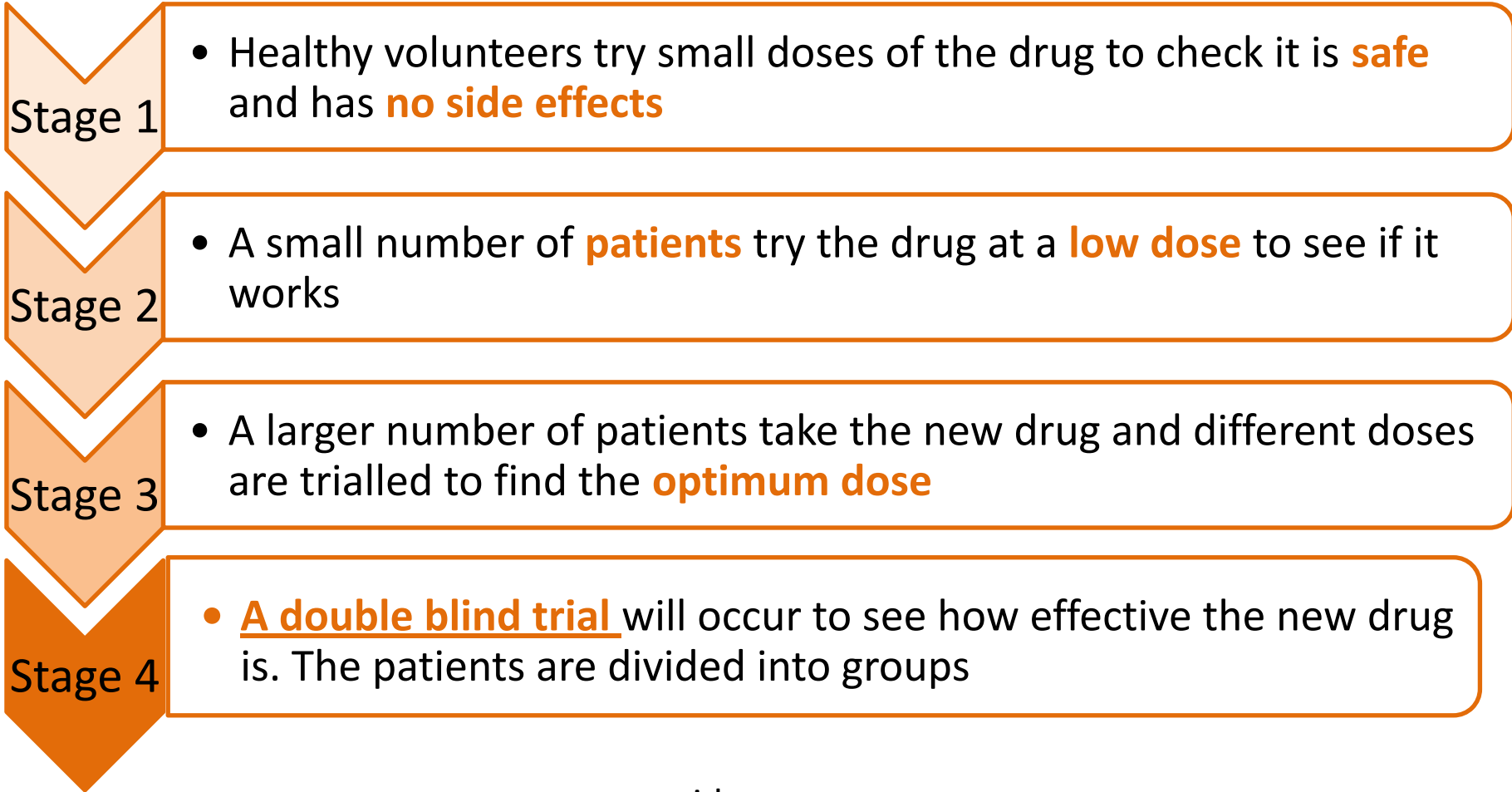


Preclinical tests must be carried out before humans are allowed to take the drug.

These preclinical tests are done on **cells**, **tissue** samples and live **animals**.

If successful the new drug will proceed to a **clinical trial**.

The stages of a clinical trial are:



video

A **double blind trial** is carried out on **patients** who **have** the **disease** to make sure that it is actually the drug which is having an effect on the patients and not anything else.

The details of everyone taking part in the trial is entered into a computer database. The computer groups people **randomly**.

Each person receives a unique code and they receive the medicine which is labelled with the matching code.

No-one knows who is receiving the real drug or a **placebo** until the end of the trial.

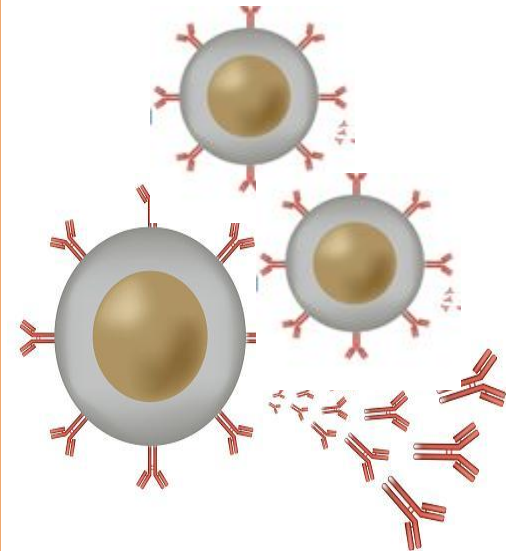
The **placebo** is often the drug the patient was originally taking so they still receive medical treatment.

Monoclonal antibodies are identical copies of **one** type of **antibody** produced in a laboratory.

How to produce monoclonal antibodies:

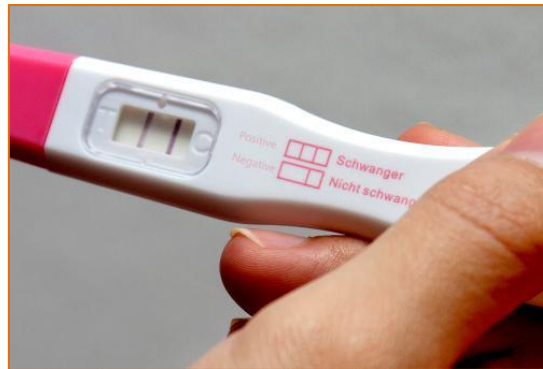
1. A mouse is **injected** with a pathogen
2. White blood cells called **lymphocytes** produce **antibodies**
3. Lymphocytes are removed from the mouse and **fused** with rapidly dividing mouse **tumour cells**
4. The new cells are called **hybridomas**.
5. The hybridomas divide rapidly and release lots of **antibodies** which are then collected.

Mono = one



Monoclonal antibodies are **specific** to one **binding site** on the **antigen**.

This means that we can use monoclonal antibodies to **target** a specific **chemical** or specific **cells** in the body.



During pregnancy, a **hormone called HCG** is released into the blood. **Monoclonal antibodies** can detect the presence of this chemical at low levels and are used in **pregnancy testing kits**.

Monoclonal antibodies are also used in treating some **cancers**.

- The monoclonal antibodies are **specific** to the antigen on the cancer cells.
- A chemical which **stops cells dividing** can be bound to the monoclonal antibodies.
- The patient is given the monoclonal antibodies and they attach to **receptors** on the cancer cells.
- The chemical **stops** the cancer cells growing and dividing.

Monoclonal antibody research has raised some concerns about their **ethical** use.

Mice are injected with pathogens and so they will **experience** the **disease** symptoms.



Mice provide antibody producing cells and tumour cells. Mice are **induced** with **cancer** to get the tumour cells.

Monoclonal antibodies have been **successful** in treating some cancers and diabetes. But there have been **deaths** when used to treat patients with **multiple sclerosis** (MS).

Herceptin is used for breast cancer. It is **not toxic** and is **specific** to breast cancer cells so **few side effects**. But it is **costly**.

QuestionIT!

Infection and Response Part 2

- Discovery of Drugs
- Development of Drugs
- Monoclonal Antibodies (Biology HT only)



1. Choose the answer from the box to complete the sentences.

Willow tree bark

Foxgloves

Laboratory

Mould

Digitalis is a drug used for heart conditions.

It originates from _____.

Aspirin is a drug used as a painkiller.

It originates from _____.

Penicillin is an antibiotic drug.

It originates from _____.

2. Any potential new drug has to go through a series of tests.

What is meant by the following terms?

a) Efficacy b) Toxicity c) Optimum dose

3. Number these statements to show how drugs are tested.

Drugs are trialled on live animals

Drugs are trialled on people with the disease the drug is for

Drugs are trialled in laboratories on cells and tissue cultures

Drugs are trialled on healthy volunteers

4. Drugs companies often use a placebo in their trials.

a. What is a placebo? b. Why is it used?

6. What are the applications of monoclonal antibodies?
7. Why do some people feel the use of monoclonal antibodies is unethical?

AnswerIT!

Infection and Response Part 2

- Discovery of Drugs
- Development of Drugs
- Monoclonal Antibodies (Biology HT only)



1. Choose the answer from the box to complete the sentences.

Willow tree bark

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Laboratory

Mould

Digitalis is a drug used for heart conditions.

It originates from **Foxgloves**.

Aspirin is a drug used as a painkiller.

It originates from **willow tree bark**.

Penicillin is an antibiotic drug.

It originates from **mould**.

2. Any potential new drug has to go through a series of tests.

What is meant by the following terms?

- a) Efficacy **how effective the drug is**
- b) Toxicity **how poisonous the drug is**
- c) Optimum dose **minimum amount of the drug which provides the best response**

3. Number these statements to show the process of drug testing.

- 2.** *Drugs are trialled on live animals*
- 4.** *Drugs are trialled on people with the disease the drug is for*
- 1.** *Drugs are trialled in laboratories on cells and tissue cultures*
- 3.** *Drugs are trialled on healthy volunteers*

4. Drug companies often use a placebo in their trials.

a. What is a placebo?

A medicine that does not contain the drug that is being trialled.

b. Why is it used?

A placebo is used to check that there are no other factors which may cause the patient's condition to improve without the drug.

6. What are the applications of monoclonal antibodies?

Pregnancy testing

Used to measure levels of hormones and other chemicals in blood

Treat some cancers by delivering a toxic drug to the cancer cells

**To find various molecules in cells or tissues by binding to them with a
fluorescent dye**

7. Why do some people feel the use of monoclonal antibodies is unethical?

Mouse has to suffer the disease deliberately

Mouse is induced to have cancer

Treatment not always safe

It's very expensive and money could be better used

LearnIT! KnowIT!

Infection and Resp Part 3

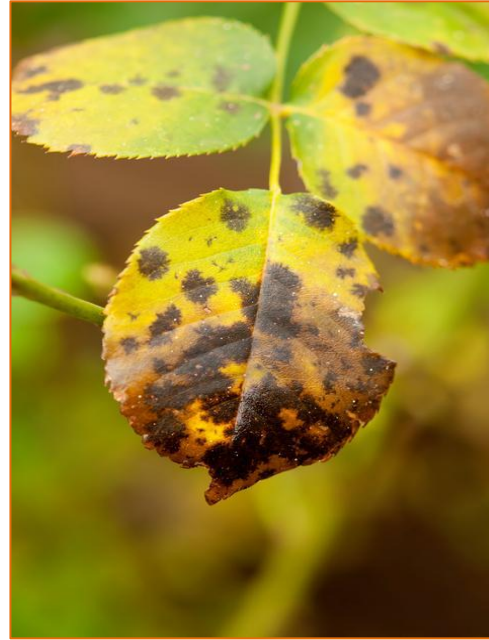
- Plant Disease (biology o



Plants suffer from a range of infections caused by pathogens or insect pests. We can detect a plant is diseased by these signs:



Unusual **growths**



Spots or discoloured
leaves



Malformed
leaves or stems



Thick waxy layers, tough cellulose cell walls and bark defend the plant from pathogen entry



Physical

video

Mechanical



Touch can make Mimosa leaves suddenly curl which frightens animals

Plants have several ways of defending themselves from pathogens and to deter herbivores.



Antibacterial and poisonous chemicals are found in many plants like witch hazel.

Chemical

Mechanical



Thorns make plants painful to be eaten

Healthy plants need **mineral ions**.

Nitrate ions are needed in **protein synthesis** and hence growth.

Horticulturists use knowledge about the effect of **ion deficiencies** to supply plants with optimum conditions.



Nitrate ions = healthy growth

Lack of nitrate = stunted growth

Magnesium = healthy chlorophyll

Lack of magnesium = chlorosis (yellowing of leaves due to lack of chlorophyll)



The presence of **pests**



Stunted growth

Areas of
decay
(rot)



If a plant disease is suspected then it may be **identified** using:

- ✓ Gardening manuals
- ✓ Gardening websites
- ✓ Test kits containing **monoclonal antibodies**
- ✓ Taking infected plants to a laboratory to identify the pathogen

QuestionIT

Infection and Respon Part 3

- Plant Disease (biology HT only)



1. Name 3 ways you can detect that a plant is diseased.
2. Copy and complete the table below:

Type of plant defence used (Mechanical, physical or chemical)	What is the plant being defended from?	Describe the defence being used
	Herbivores eating it	Thorns or hairs
Chemical		The chemical released is antibacterial
	Herbivores and pathogen entry	Dead bark coating which falls off
Physical	Insects such as aphids	

1. Name 3 ways you can detect that a plant is diseased.

leaf spots/discolouration, growths, malformed stems or leaves, presence of pests, stunted growth, areas of decay (rot)

2.

Type of plant defence used (mechanical, physical or chemical)	What is the plant being defended against?	Describe the defence being used
Mechanical	Herbivores eating it	Thorns or hairs
Chemical	Pathogens/bacteria Herbivores/animals	The chemical released is antibacterial or poisonous
Physical	Herbivores and pathogen entry	Dead bark coating which falls off
Physical	Insects such as aphids	Waxy cuticle/cellulose cell walls are hard to penetrate

3. Describe why nitrate ions are very important for plants.
4. What is chlorosis and how can it be prevented?
5. Why would a test kit containing monoclonal antibodies be useful to detect plant disease?

3. Describe why nitrate ions are very important for plants.
Nitrate is crucial for protein synthesis and hence growth. Without sufficient nitrate ions, the plant would be stunted.
4. What is chlorosis and how can it be prevented?
Chlorosis is the yellowing of the leaves. It can be prevented by providing the plant with magnesium.
5. Why would a test kit containing monoclonal antibodies be useful to detect plant disease?
The monoclonal antibodies can be used to detect a particular chemical which is only present in one type of pathogen. This would then show whether the plant was infected with this disease or not.

Questions for Biology HT only

5. Place the statements in the correct order to show how monoclonal antibodies are produced.

- ☐ Mouse lymphocytes produce antibodies.
- ☐ The antibodies are collected and purified.
- ☐ Mouse lymphocytes are collected.
- ☐ Mouse is injected with a pathogen.
- ☐ Hybridoma cells are separated and cultured to form a clone.
- ☐ Mouse lymphocyte and mouse tumour cell are fused.
- ☐ The clone produces large quantities of antibody.

Questions for Biology HT only

5. Place the statements in the correct order to show how monoclonal antibodies are produced.
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 5. Hybridoma cells are separated and cultured to form a clone.
 4. Mouse lymphocyte and mouse tumour cell are fused.
 6. The clone produces large quantities of antibody.

AnswerIT!

Infection and Response Part 3

- Plant disease (biology only)

